Clinic document retrieval

Issue #5 | Work summary | 2 November 2026

Completed and pushed. A reproducible comparison of keyword search and PostgreSQL full-text search, with source-preserving results and a proposed architecture decision.

What was delivered

Experiment: Five synthetic clinic passages and 12 questions, including supported questions, paraphrases, and unsupported requests. Both methods return up to three passages.
Evidence: Actual results preserve passage IDs, original source names, text, scores, misses, and irrelevant hits.
Documentation: Research findings, official references, repeatable commands, and an ADR marked **Proposed**.

Key results

Measure	Keyword	PostgreSQL
Supported questions with expected evidence in top 3	**7 / 8**	**6 / 8**
Unsupported questions returning any evidence	**2 / 4**	**0 / 4**
Total irrelevant passages returned	**11**	**0**

Important findings and recommendation

Propose PostgreSQL as a conservative starting point: it returned cleaner evidence in this small sample, but found fewer supported questions. Both methods missed the car/parking paraphrase. PostgreSQL also missed the parking-cost question because every remaining query term must match.
A retrieved passage does not prove an answer. Preserve source IDs throughout the workflow, verify the specific fact, and abstain or refer to staff when evidence is missing or insufficient. The two recipes use different matching and stop-word rules; this is not an isolated test of stemming.

Verification and current status

Verified: Four unit tests passed. Two PostgreSQL 18.6 runs produced identical results; retrieved text and source IDs matched the fixtures. Git whitespace checks passed. The temporary database was stopped.
Limitations: Synthetic English data only; no production answer workflow or scale benchmark. The Docker Compose path was provided but not executed. **Team agreement is required before accepting the ADR.**

Delivery details

Branch: feature/5-hafisaalbahri-clinic-document-retrieval
Commit: dc63995 - Compare clinic document retrieval approaches for issue #5
Files: experiments/retrieval/; docs/research/clinic-document-retrieval.md; docs/adr/0001-clinic-document-retrieval.md; README.md

GitHub issue #5 | View implementation commit